



Heat transfer augmentation in a microchannel heat sink with sinusoidal cavities and rectangular ribs



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ABSTRACT

A three-dimensional numerical simulation was conducted to study the characteristics of fluid flow and heat transfer in new design of microchannel heat sink with sinusoidal cavities and rectangular ribs (MC-SCRR) for Reynolds number ranging from 100 to 800. A comparative analysis has conducted to the performance of the proposed design with related geometries such as microchannel with rectangular ribs MC-RR and microchannel with sinusoidal cavities MC-SC. The results showed that thermal performance of MC-SCRR is superior over both MC-RR and MC-SC. The new design of MC-SCRR has proved the ability to combine between two important features; large flow area which significantly reduces the pressure drop and high flow disturbances which caused by existence of ribs in the central portion of channel. The overall performance of MC-SCRR is evaluated in term of friction factor, Nusselt number and performance factor. The effect of three geometrical parameters; relative cavity amplitude (λ) relative rib width (β) and relative rib length (Γ) on the convective heat transfer and pressure drop have been investigated. The performance factor Pf for MC-SCRR with $\lambda = 0.15$, $\beta = 0.13$ and $\Gamma = 0.5$ achieves 1.85 at $Re = 800$.

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1. Introduction

The rapid evolution in Integrated circuits has been associated with a considerable increase in generated heat flux as a consequence of high circuits dense in a miniature size. This led to a rising demand for more effective cooling method than the traditional methods of air-cooling. The fast development in micro-electro mechanical systems (MEMS) has motivated the researchers to be utilized in developing of new micro-cooling techniques. A lot of techniques have been studied such as; micro-jet impingement, micro-heat pipe, micro-electro-hydrodynamic and microchannel heat sink. Within these techniques, the microchannel heat sink has shown high efficiency when Tuckerman and Pease [1] have investigated experimentally heat transfer in silicon microchannel heat sink. The study emphasized the ability of microchannel heat sink to remove heat rate up to 790 W/cm^2 . The high surface area to volume which provides by microchannel has significantly contributed in increasing the thermal efficiency. Additionally, the highly thermal conductivity and specific heat for liquids have also led to enhance heat transfer performance.

Since the invention of microchannel heat sink, a lot of numerical and experimental studies have been dedicated to investigate the characteristics of heat transfer and fluid flow in straight rectangular microchannel [2–4]. There are many restrictions that hinder the improvement of thermal performance in straight simple microchannel. The high pressure drop across the microchannel is one of the most significant limitations because it contributing in increasing pumping power consumption and leakage risk. Additionally, the micro size of the channels makes the flow invariably in laminar region, thus leads to low performance compared with turbulent flow. As consequences of the continuous increase in heat loaded and rigorous requirement of temperature determination for electronic elements, the straight simple channel is unable to satisfy these demands. So, a lot of studies have diverted the interest towards passive techniques which applied in conventional channels to be utilized in microchannel. Steinke and Kandlikar [5] proposed several techniques which are enforceable in microchannels to enhance heat transfer in microchannel such as incorporation of mixing features to improve the mixing flow, increase local heat transfer coefficient by breaking boundary layer utilizing intermittent construction, incorporation of grooves and ribs. Accordingly, many studies have adopted these strategies to improve the thermal performance of microchannel heat sink. Channel curvature technique has proved effectiveness in promoting heat transfer

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through many experimental and numerical studies. The effect of centrifugal force due to the curvature leads to emergence the Dean vortices and chaotic advection at the troughs and crests which are significantly enhance flow mixing. Sui et al. [6] investigated numerically heat transfer enhancement in sinusoidal wavy microchannel. They highlighted the effect of increasing the waviness in flow direction and locally at specific region to remove heat flux. The study showed high heat transfer performance factor of 1.71–2.95 with reasonable pressure drop penalty factor of 1.38–2. Sui et al. [7] have conducted an experimental study using wavy microchannel with rectangular cross-section. The results showed a progressive in heat transfer performance with a ratio of 211% comparative with simple channel, whereas the pressure drop penalty has witnessed a smaller increment of 76%. Xie, et al. [8,9] performed numerically a comparative study between longitudinal and transverse wavy microchannel. They revealed the superiority of transverse wavy microchannel over longitudinal in terms of lower pressure drop. Mohammed H.A. et al. [10], Dai et al. [11] and Dai et al. [12] have asserted through comparative studies the superiority of zigzag microchannel over sinusoidal microchannel in heat transfer performance at the expense of higher pressure drop. Besides, the periodic convergent-divergent construction has also showed effectiveness in promoting heat transfer. Ghaedamini et al. [13] carried out numerically the effect of geometrical parameters on hydrothermal performance in convergent-divergent microchannel. Their analysis emphasized that increasing the waviness has produced chaotic advection which increase the flow mixing and therefore enhance heat transfer

The flow disruptions techniques have also attracted interest as an efficient passive heat transfer augmentation method through using offset strip fins, interrupted-wall channel, grooves, cavities and ribs.

Xu et al. [14,15] studied experimentally and numerically the effect of the periodic interruption of flow in longitudinal microchannels with transverse microchambers on heat transfer enhancement. They have revealed that interruption and redevelopment of thermal boundary layer at the leading edge of each section has a significant effect on improving heat transfer. The results showed that substrate temperature has been reduced by 14 °C less than simple straight channel. Additionally, the pressure drop has also reduced due to shorten flow path.

Chai et al. [16] proposed of using rectangular ribs in transverse microchambers between interrupted microchannels. They emphasized experimentally and numerically that overall performance showed an enhancement with Reynolds number less than 600 due to the dominance of heat transfer enhancement over pressure drop penalty. The study has attributed this enhancement to the effect of ribs which contribute in the occurrence of flow separation, vortices, recirculation and boundary layer separation.

Cavities and grooves are also widely used to enhance heat transfer due to the effect of jet and throttling of flow which increase the mixing between fluid layers near the wall and central core. Moreover, the emergence of swirls inside cavities due to pressure gradient produces further flow mixing.

Xia et al. [17–21] studied numerically and experimentally the effects of several shapes of re-entrant cavities on the hydrothermal performance of microchannel. These cavities include; aligned fan-shaped re-entrant cavities, offset fan-shaped re-entrant cavities, triangular re-entrant cavities. All studies have asserted the superiority of microchannel with cavities over simple straight microchannel. The configuration of microchannel with triangular re-entrant cavities has shown highest thermal performance at expense of pressure drop penalty.

Abouali and Baghernezhad [33] carried out a comparative study of thermal performance of arc grooved and rectangular grooved microchannels. The grooves have been created on the floor of

microchannel besides their existence on side-walls. The study revealed that thermal performance of arc grooved microchannel is superior over rectangular grooved microchannel. In comparison with simple microchannel, the overall performance factor has witnessed increase of 4.4% and 14% for rectangular grooved and arc grooved respectively.

Kuppusamy et al. [22,23] investigated numerically the effect of geometrical parameters in trapezoidal and triangular grooves. They showed that triangular grooves performance is higher than trapezoidal. They pointed out to the role of increasing the angle and the depth in expanding the groove zone which give a space to form larger vortices that enhance flow mixing. Also, Ahmed et al. [24] investigated numerically the impact of geometrical parameters on heat transfer and fluid flow in three types of grooves; triangular, trapezoidal and rectangular. These parameters comprise; depth, tip length, pitch and orientation ratio. The results emphasized that trapezoidal grooves, with optimal geometrical parameters, have shown better thermal performance than triangular at Reynolds number of 100. The Nusselt number has increased about 51.59% with minor increase in the friction factor of 2.35%.

Ribs are another structure which used to enhance heat transfer through breaking thermal and hydraulic boundary layers. Furthermore, the ribs can promote flow mixing through vortices creation and chaotic advection. However, the heat transfer enhancement by ribs is accompanied with high pressure drop penalty. So, it requires an optimization procedure to the ribs construction in order to maintain lower pressure drop. Chai et al. [25–27] studied the effect of fan-shaped ribs on hydrothermal performance of microchannel heat sink with regards to geometrical parameters. Two types of ribs arrangement have been studied; aligned and offset. The results have emphasized that ribs with lower height and larger spacing resulting in better heat transfer performance. Additionally, it has been revealed that microchannel with align ribs arrangement is thermally more efficient than offset arrangement. The Nusselt number has increased about 6–101% and 4–103% for align ribs arrangement and offset ribs arrangement respectively. At the same time, the friction factor has increased significantly about 1.1–8.28 and 1.22–6.27 for align ribs arrangement and offset ribs arrangement respectively.

Recently, Chai et al. [28] studied the effects of five different ribs shapes on the hydrothermal performance of microchannel heat sink. These shapes comprise; rectangular, forward triangular, backward triangular, isosceles triangular, and semi-circular. The results indicated that microchannel with forward triangular offset ribs showed highest performance in Reynolds number less than 350. At the same time, microchannel with semi-circular ribs exhibited the highest performance with Reynolds number >400. The overall performance is ranged between (1.02–1.48) in comparison with simple microchannel.

The hybrid strategy between ribs and either grooves or cavities has also adopted as an effective technique to enhance heat transfer. Xia et al. [34] investigated the hydrothermal characteristics of microchannel heat sink with circular ribs and fan-shaped re-entrant cavities. The ribs are installed in straight space between cavities. The study has emphasized that combining ribs with cavities have a significant effect on increase heat transfer. The study attributed this enhancement to the emergence of vortices due to periodic pressure variation which leads to increase flow mixing. Furthermore, the periodic interruption and redevelopment to thermal boundary layer has also contributed in heat transfer enhancement. The results analysis showed that the moderate relative rib height of $e/Dh = 0.12$ at Reynolds number of (592), has achieved the highest thermal enhancement of (1.6). Recently, Li et al. [29] have followed a new strategy of using triangular grooves arranged on side-walls with ribs in central portion of the channel. This configuration proved the ability to enhance heat transfer because it

provides more heat transfer area, promotes jetting and throttling and contributes in a periodic interrupting and redeveloping to the hydraulic thermal boundary layer. Besides, the increasing in flow area reduces the pressure drop that caused by ribs.

It is obviously from above literature review, many studies have been done to optimize the geometric structure using a combination between ribs and cavities with different geometries such as triangular, trapezoidal, and semi-circular. But there is a need to explore new designs that provides higher thermal performance in a lower pressure drop. The present study aims to utilize the sinusoidal cavities arranged on the side walls of microchannel with ribs in the central portion of the channel to enhance the performance of MCHS. Using cavities with relatively long wave length of 0.5 mm, the number of cavities and ribs are reduced such that to be less complexity in fabrication process. Besides, it offers larger flow area which significantly reduces pressure drop. The combined effects of sinusoidal cavities and rectangular ribs on the hydrothermal performance of MCHS are studied numerically. Furthermore, an optimization process has been applied to find optimal geometrical parameters of cavities and ribs which provide best performance.

2. Mathematical modelling of microchannel heat sink

The microchannel heat sink consists of ten microchannels made of silicon. In order to save the computational cost, only one symmetrical part of MCHS is adopted in present simulation. Figs. 1 and 2 illustrates the computational domain with geometrical parameters for both of sinusoidal cavities and rectangular ribs (MC-SCRR) and corresponding straight rectangular microchannel MC-RC. As shown in Fig. 1, the main dimensions of MCHS such as total length L_t , total width W_t and total height H_t is 10 mm, 0.3 mm and 0.4 mm respectively. The height of microchannel H_c is 0.3 mm and the width of the channel's wall which separates between two channels is 0.15 mm. All of the geometrical parameters

are shown in Table 1. In order to study the impact of sinusoidal cavities and rectangular ribs on the fluid flow and heat transfer, three geometrical parameters are selected to be changed; the relative cavity amplitude (λ), the relative rib width (β) and the relative rib length (Γ). The relative cavity amplitude is defined as the ratio of Amplitude (A) to the cavity length (L_c) and the relative rib width (β) is defined as the ratio of rib width (W_r) to the channel width (W_c). The relative rib length is defined as the ratio of rib length (L_r) to the cavity length (L_c). The relative cavity amplitude (λ) is changed to 0.05, 0.1, 0.15, 0.2, the relative rib width (β) is changed to 0.2, 0.3, 0.4, 0.5, 0.6, while the relative rib length is changed to 0.4, 0.5, 0.6, 0.7, 0.8 respectively.

3. Numerical method

The computational fluid dynamic (CFD) software FLUENT 14 is used to solve the three dimensional heat transfer and fluid flow equations according to the following assumptions.

- According to [30], the flow is considered continuum if the Knudsen number (Kn) is less than (10^{-3}). In present study, the (Kn) is far less than (10^{-3}), so the flow could be assumed continuum and the Navier-Stokes equation and non-slip boundary condition are applicable.
- The fluid is Newtonian, incompressible and the flow is laminar.
- Both fluid flow and heat transfer are in steady-state.
- Constant thermophysical properties.
- The viscous dissipation and gravitational force are neglected.
- The radiation heat transfer is neglected.

3.1. Governing equations

According to the previous assumptions, the governing equations of mass, momentum and energy can be written as follows:

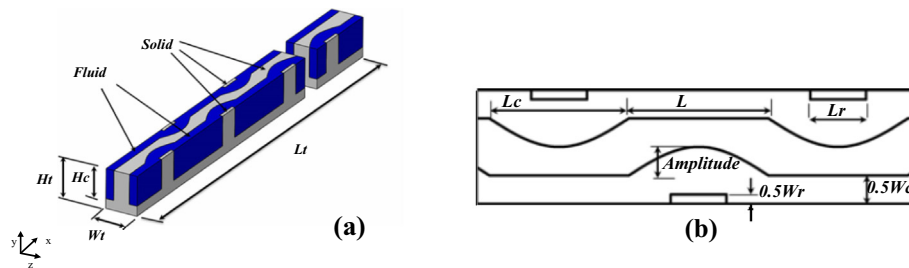


Fig. 1. (a) Schematic diagrams of MC-SCRR (b) geometric parameters of MC-SCRR.

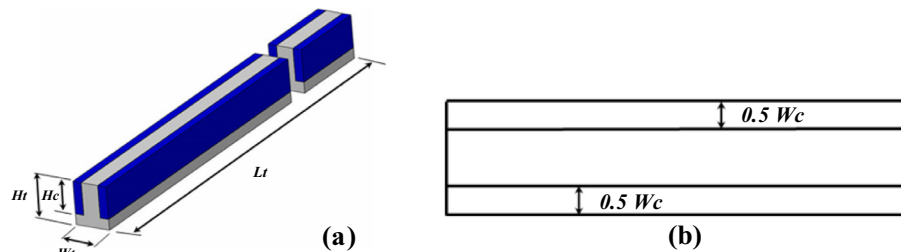


Fig. 2. (a) Schematic diagrams of MC-RC (b) geometric parameters of MC-RC.

Table 1

The geometrical parameters of SC-RR MCHS.

Geometrical parameters	Amplitude (A)	Hc	Ht	Lt	Lc	L	Lr	Wc	Wr	Wt
Value/ μm	25, 50, 75, 100	300	400	10,000	500	500	200, 250, 300, 350, 400	150	30, 45, 60, 75, 90	300

Continuity equation:

$$\frac{\partial u}{\partial x} + \frac{\partial v}{\partial y} + \frac{\partial w}{\partial z} = 0 \quad (1)$$

where u, v and w are the velocity components in x, y and z -directions respectively.

Momentum equations:

$$u \frac{\partial u}{\partial x} + v \frac{\partial u}{\partial y} + w \frac{\partial u}{\partial z} = -\frac{1}{\rho_f} \frac{\partial p}{\partial x} + \frac{\mu_f}{\rho_f} \left(\frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2} + \frac{\partial^2 u}{\partial z^2} \right) \quad (2)$$

$$u \frac{\partial v}{\partial x} + v \frac{\partial v}{\partial y} + w \frac{\partial v}{\partial z} = -\frac{1}{\rho_f} \frac{\partial p}{\partial y} + \frac{\mu_f}{\rho_f} \left(\frac{\partial^2 v}{\partial x^2} + \frac{\partial^2 v}{\partial y^2} + \frac{\partial^2 v}{\partial z^2} \right) \quad (3)$$

$$u \frac{\partial w}{\partial x} + v \frac{\partial w}{\partial y} + w \frac{\partial w}{\partial z} = -\frac{1}{\rho_f} \frac{\partial p}{\partial z} + \frac{\mu_f}{\rho_f} \left(\frac{\partial^2 w}{\partial x^2} + \frac{\partial^2 w}{\partial y^2} + \frac{\partial^2 w}{\partial z^2} \right) \quad (4)$$

where ρ_f and μ_f are the density and dynamic viscosity of the coolant, respectively, and p is the coolant pressure.

Energy equation for the coolant

$$u \frac{\partial T_f}{\partial x} + v \frac{\partial T_f}{\partial y} + w \frac{\partial T_f}{\partial z} = \frac{k_f}{\rho_f c_{pf}} \left(\frac{\partial^2 T_f}{\partial x^2} + \frac{\partial^2 T_f}{\partial y^2} + \frac{\partial^2 T_f}{\partial z^2} \right) \quad (5)$$

where T_f is the coolant's temperature, c_{pf} is fluid specific heat and k_f is fluid thermal conductivity.

Energy equation for the solid region:

$$0 = k_s \left(\frac{\partial^2 T_s}{\partial x^2} + \frac{\partial^2 T_s}{\partial y^2} + \frac{\partial^2 T_s}{\partial z^2} \right) \quad (6)$$

where T_s is solid temperature and k_s is solid thermal conductivity

Boundary condition

1- channel inlet at $x = 0$

$u = u_f, v = w = 0$

For fluid $T_f = T_{in} = 300$ K

For solid $-k_s \left(\frac{\partial T_s}{\partial x} \right) = 0$

2- channel outlet at $x = 10$ mm

$p_f = p_{out} = 1$ atm

For fluid $-k_f \left(\frac{\partial T_f}{\partial x} \right) = 0$

For solid $-k_s \left(\frac{\partial T_s}{\partial x} \right) = 0$

3- upper wall at $y = 0.4$ mm

$u = v = w = 0$

$-k_s \left(\frac{\partial T_s}{\partial y} \right) = 0$

4- Inner walls of microchannel (fluid / solid surface)

$u = v = w = 0$

$-k_s \left(\frac{\partial T_s}{\partial n} \right) = -k_f \left(\frac{\partial T_f}{\partial n} \right)$ Where (n) is the coordinate normal to the wall

5- At $z = -0.15$ mm $\partial/\partial z = 0$ (symmetry)

At $z = 0.15$ mm $\partial/\partial z = 0$ (symmetry)

6- At lower wall (substrate) $y = 0$

$-k_s \left(\frac{\partial T_s}{\partial y} \right) = q = 100$ W/cm²

3.2. Grid independence and CFD simulation

The (ICEM) software is used to generate mesh for the solver. Fig. 3a shows the mesh structure in x - z plane ($y = 0.25$ mm) of the (SC-RR) MCHS with $\beta = 0.333$, $\lambda = 0.15$ and $\Gamma = 0.4$. The structured mesh is applied in all zones with uniform grids. Finite volume method is utilized to discretize the governing equation. The SIMPLE algorithm was adopted to accomplish the pressure-velocity coupling. At the same time, the second order upwind scheme is used for convective term and second order central difference scheme is applied for diffusion term. Furthermore, convergence criterions are set to be less than 10^{-6} for continuity and less than 10^{-9} for energy.

A grid independence test is conducted to find the optimum mesh structure which produce accurate results in lower time cost. The grid number of (SC-RR) MCHS with $\beta = 0.333$, $\lambda = 0.15$ and $\Gamma = 0.4$ is varied from 0.75 million to 1.4 million for Reynolds number of 600. The relative error is calculated according to following equation:

$$e\% = \left| \frac{J_2 - J_1}{J_1} \right| \times 100 \quad (7)$$

where (J) represents any parameter; such as temperature, Nusselt number and pressure drop, J_1 is represents the value of parameter

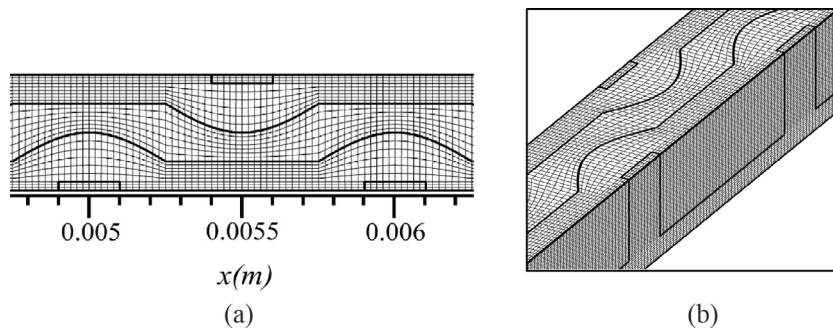


Fig. 3. (a) Computational grid in the x - z plane of MC-SCRR with $\beta = 0.333$, $\lambda = 0.15$ and $\Gamma = 0.4$ (b) Three dimensional geometry of MC-SCRR with mesh.

Table 2
Grid independents test.

No.	Grid numbers/ $\times 10^6$	Nu	$e\%$	Pressure drop/Pa	$e\%$
1	1.4	19.66	–	54953.5	–
2	1.235	19.652136	0.04	54678.7325	0.5
3	1.021	19.646238	0.07	54458.9185	0.9
4	0.855	19.64034	0.1	54074.244	1.6
5	0.623	19.634442	0.13	53799.4765	2.1

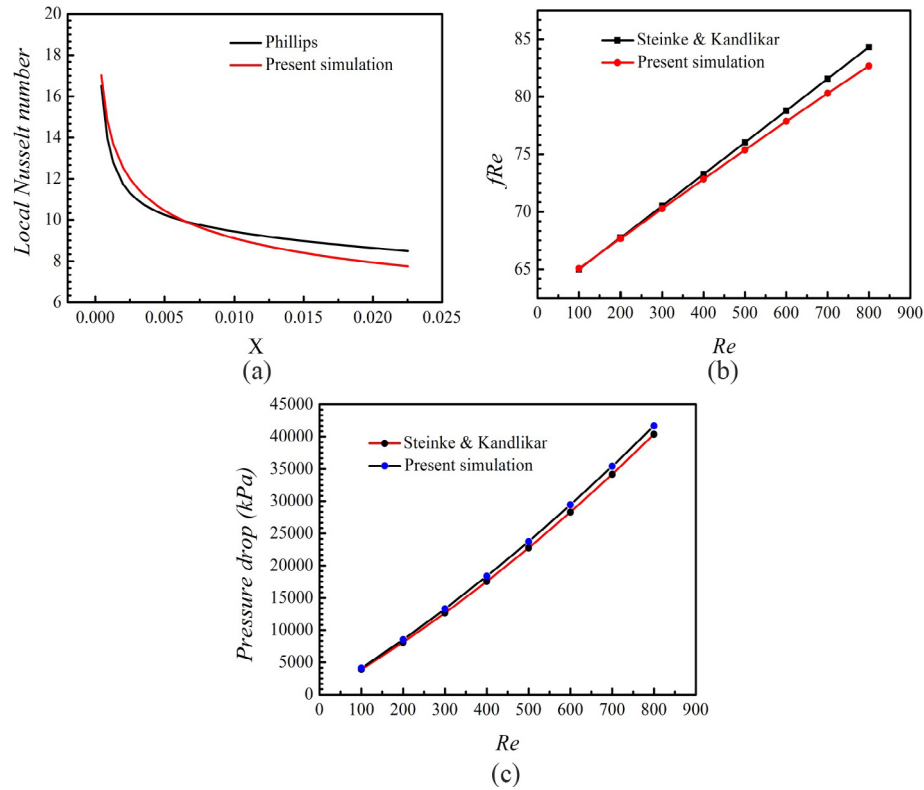


Fig. 4. Numerical validation (a) local Nusselt number according to Phillips [32] (b) Poiseuille number, and (c) Pressure drop according to Steinke & Kandlikar [31].

which acquired from finest grids, whereas J_2 represents the value of parameters obtained from other grids. Table 2 shows the relative errors for pressure drop and Nusselt number of different grids comparative with the finest grids of 1.4 million. It is clearly observed that grid number of 1.021 million has a reasonable accuracy compared with other grid numbers. Consequently, a grid of 1.021 million cells has been used in current study.

3.3. Data reduction

This section presents the relevant expressions that used to calculate the characteristics of heat transfer and fluid flow in MCHS. The Reynolds number (Re) is expressed as follow:

$$Re = \frac{\rho u_m D_h}{\mu} \quad (8)$$

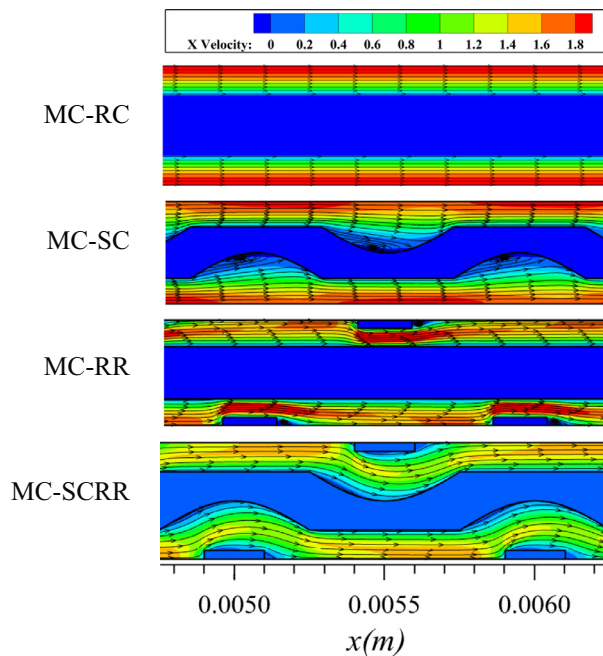


Fig. 5. streamline and velocity distribution for various structures on x-z plane ($y = 0.25$ mm) at $Re = 200$.

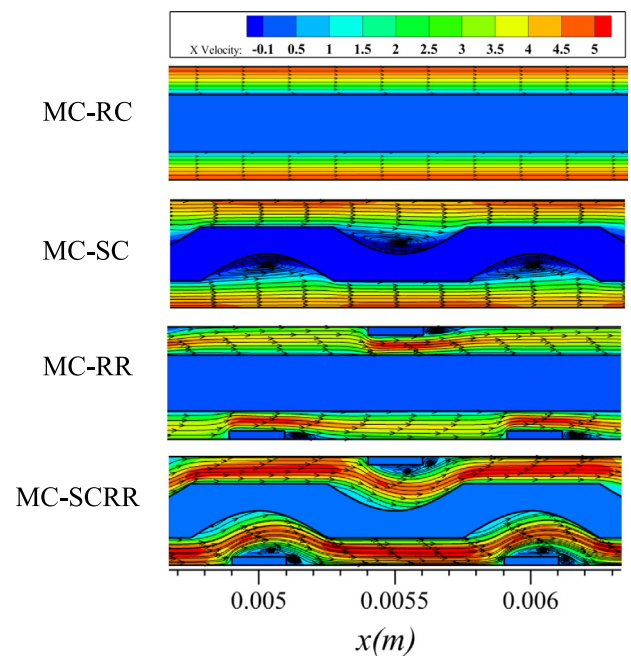


Fig. 6. streamline and velocity distribution for various structures on x-z plane ($y = 0.25$ mm) at $Re = 500$.

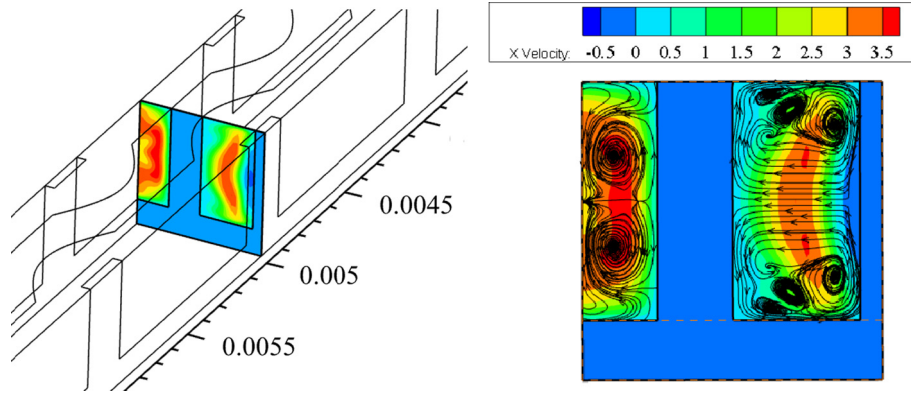


Fig. 7. streamlines and velocity distribution in z-y plane at x = 5 mm describes the transverse flow motion towards cavity depth.

where D_h represents the hydraulic diameter and it is calculated as follow:

$$D_h = \frac{2H_c W_c}{H_c + W_c} \quad (9)$$

where ρ , u_m , μ , H_c and W_c are fluid density, mean velocity, dynamic viscosity, height of microchannel and width of microchannel respectively.

The apparent friction factor in rectangular duct is given by

$$f_{app} = \frac{2D_h \Delta P}{L_t \rho u_m^2} \quad (10)$$

where L_t and ΔP are the total length of microchannel and pressure drop across microchannel respectively.

The average heat transfer coefficient is given by

$$h_{ave} = \frac{q_w A_{film}}{A_{con.} (T_{w,ave} - T_{f,ave})} \quad (11)$$

where q_w , A_{film} , $A_{con.}$, $T_{w,ave}$ and $T_{f,ave}$ are the heat flux per unit area, heated area, convection heat transfer area, average temperature of microchannel wall and average fluid temperature.

The average Nusselt number is given by

$$Nu_{ave} = \frac{h_{ave} D_h}{k_f} \quad (12)$$

where k_f is the thermal conductivity of the fluid.

4. Results and discussion

4.1. Validation of rectangular microchannel (RC) based on the conventional predictions

In order to verify the accuracy of present numerical simulation, the Nusselt number, pressure drop and Poiseuille number (P_o) for the rectangular microchannel model (MC-RC) are validated with theoretical results. The correlations of Steinke & Kandlikar [31] and Philips [32] have been used to acquire the theoretical results of local Nusselt number and friction factor respectively.

The local Nusselt number of thermally developing region according to Ref. [32] can be written as:

$$Nu_X = 1.0958 \left[\frac{23.315 + 27038(X) + 1783300(X)^2}{1 + 3049(X) + 472520(X)^2 - 35714(X)^3} \right] \quad (13)$$

$$X = \frac{x}{D_h Re Pr} \quad (14)$$

The friction factor correlation according to Ref. [31] is given by

$$f_{app} = f + \frac{\alpha(x) D_h}{L_t} \quad (15)$$

The Poiseuille number (P_o) is written as

$$P_o = f Re = 96[1 - 1.3553(AR) + 1.946(AR)^2 - 1.7012(AR)^3 + 0.9564(AR)^4 - 0.2537(AR)^5] \quad (16)$$

$$\alpha(x) = 0.6796 + 1.2197(AR) + 3.3089(AR)^2 - 9.5921(AR)^3 + 8.9089(AR)^4 - 2.9959(AR)^5 \quad (17)$$

From Eqs. (15), (16) and (17), the pressure drop equation can be extracted as follow:

$$\Delta p = \frac{2(f Re) \mu u_m L_t}{D_h} + \frac{\alpha(x) \rho u_m^2}{2} \quad (18)$$

The comparisons between the theoretical results and the present simulation results are illustrated in Fig. 4. The theoretical data of local Nusselt number, Poiseuille number (P_o) and pressure drop are obtained from Eqs. (13), (16), (18) respectively. It can be obvi-

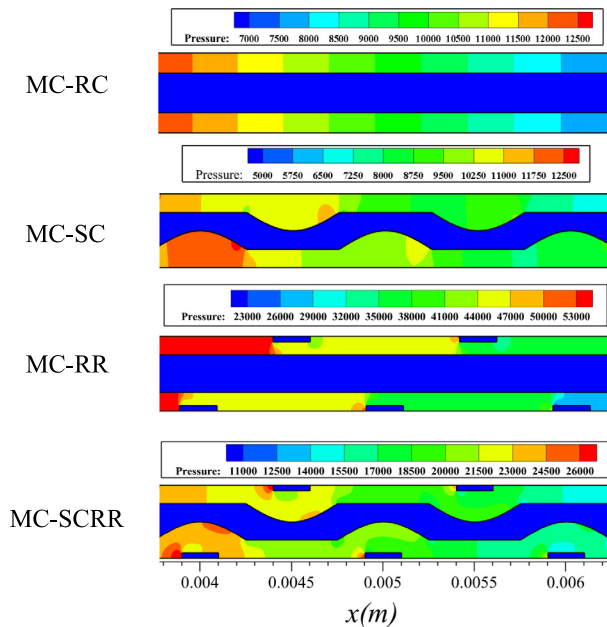


Fig. 8. The pressure distribution contours for the four types of MC's on x-z plane at y = 0.25 mm with Re = 500.

ously shown that the present numerical simulation results are in good agreement with theoretical results. This agreement indicates that the current numerical simulation method can be applied to predict the characteristics of heat transfer and fluid flow in MC-SCRR.

4.2. Fluid flow and heat transfer characteristics of microchannel with sinusoidal cavities and rectangular ribs MC-SCRR

4.2.1. Characteristics of velocity distribution

In order to recognize the effects of geometric construction of microchannel with sinusoidal cavities and rectangular ribs (MC-SCRR) on the hydro-thermal performance of MCHS, a comparison has been conducted with other relevant constructions. These constructions include; microchannel with sinusoidal cavities MC-SC, microchannel with rectangular ribs MC-RR and straight rectangular microchannel MC-RC. Figs. 5 and 6 illustrates this comparison with specific geometrical parameters of ($\beta = 0.333$, $\lambda = 0.15$ and $\Gamma = 0.4$) for MC-SCRR, ($\beta = 0$, $\lambda = 0.15$ and $\Gamma = 0$) for MC-SC and ($\beta = 0.333$, $\lambda = 0$ and $\Gamma = 0.4$) for MC-RR. The velocity and streamline distributions are given in the (x - z) plane ($y = 0.25$) at $Re = 200$ and $Re = 500$ in Figs. 5 and 6 respectively. The combination of ribs and cavities has an obvious effect on flow behaviour in microchannel comparative with other structures. At low Reynolds number of 200, Fig. 5 illustrates the role of ribs in guiding the flow trajectory. In MC-SCRR, it is clearly shown that the ribs are forcing the flow streamlines periodically to re-entrant the cavities. But in case of ribs absence, such as in MC-SC, the flow streamlines witnessed a slight deviation towards cavities which in turn forms vortexes due to the wide stagnation zone. In MC-RR, the flow is deflected by the ribs to the sidewalls without causing any vortexes. The characteristic of flow streamlines in MC-RC is the parallelism with each other without any deviations or vortexes. Regarding to the velocity distribution, the maximum velocity in both of MC-RC and MC-SC are concentrated in the central portion of microchannel, while the minimum flow velocity is found in the cavities region of MC-SC. Conversely with flow velocity distribution for the MC-RR, where the maximum velocity is focused in the sidewalls of the ribs and minimum velocity behind the ribs. In MC-SCRR, The flow accelerated in the entrance of cavities due to sudden expansion

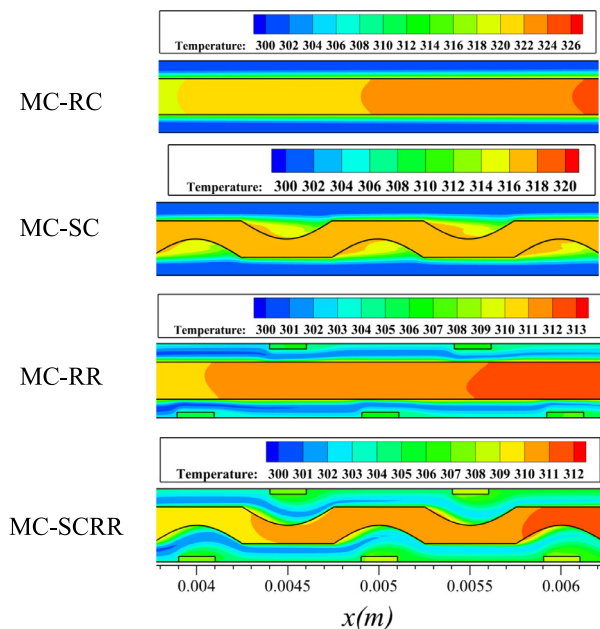


Fig. 9. The temperature distribution contours for the four types of MC's on x - z plane at $y = 0.25$ mm with $Re = 500$.

of the flow area and the flow-obstructing effect of the rib. The minimum flow velocity is observed in the downstream region of the ribs. At the same time, it has been noticed the disappearance of vortexes in cavities due to the occupation of the whole cavity region by fluid which eliminates the stagnation zone.

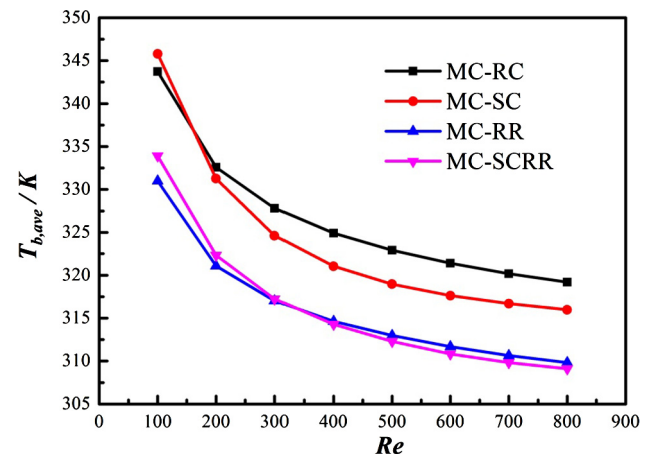


Fig. 10. The average temperature of the bottom surface of MC-RC, MC-SC, MC-RR and MC-SCRR.

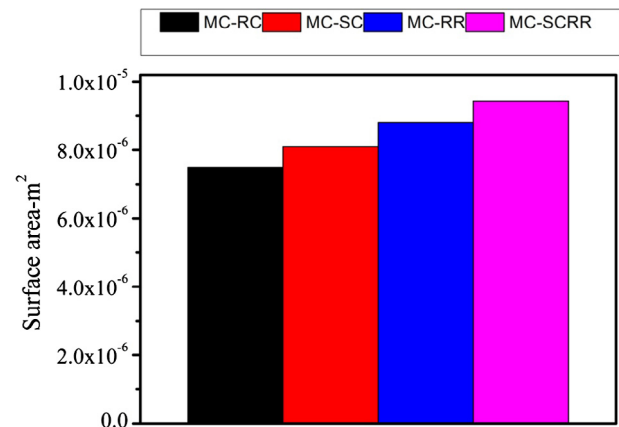


Fig. 11. The surface area of the four types of MC's.

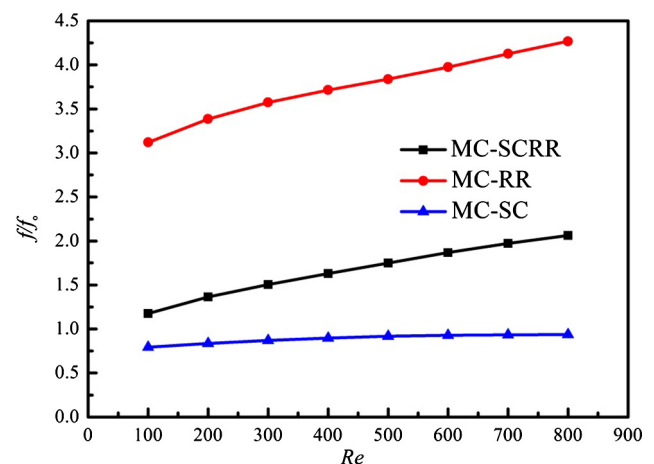


Fig. 12. Variation of friction factor ratio with Reynolds number for MC-SCRR, MC-RR and MC-SC.

At higher Reynolds number ($Re = 500$), Fig. 6 shows the same velocity distributions for MC-SC with enlargement in recirculation region inside cavities. For MC-RR, the vortices are formed in the rear zone of ribs due to high flow disturbance. In MC-SCRR, the vortices are intensified on sidewalls and rear zone of the ribs. Fig. 7 depicts more clearly the transverse flow motion towards cavity depth at y - z plane ($x = 5$ mm) in the confined region between the cavity crest and rib surface. It is showing clearly the vortices formation as a result of flow stretching and folding which significantly enhance flow mixing and chaotic advection. Besides, the vortices are intensified on sidewalls and rear zone of the ribs.

According to previous comparison, the importance of combination between sinusoidal cavity and rectangular ribs can be manifested in many aspects. The proposed design has an ability to eliminate the stagnation region which seriously impedes heat transfer as in case of MC-SC. Also, it provides a better velocity distribution due to the presence of a wide flow area. Besides, it improves flow mixing due to the vortices formation in transverse direction of cavity. The simultaneous existence of ribs and cavities leads to high rate of periodic interruption to the hydraulic boundary layer.

4.2.2. Characteristics of pressure distribution

The pressure distribution in the four microchannel; MC-RC, MC-SC, MC-RR and MC-SCRR are shown in Fig. 8. In MC-RC, the static pressure decreases along flow stream due to the friction losses. For MC-SC, the static pressure decreases in the entrance region of the cavity and increases in the contraction region. Conversely in MC-RR, the static pressure increases in the upstream of the ribs due to the flow-blocking effect, while the downstream region witnessed a lower static pressure region. The MC-RR is characterized with a higher pressure drop among the other MC's designs due to the existence of the ribs in the central portion of the channel. Compared with MC-RR, the proposed design of MC-SCRR is considerably reduces the static pressure due to the availability of larger flow area. Additionally, the pressure gradient towards transverse direction of cavity contributes in enhancing flow mixing through vortices formation as shown in Fig. 7.

4.2.3. Characteristics of temperature distribution

The temperature distribution for all cases is shown in Fig. 9. In MC-RC, it can be observed two distinct regions of temperature distribution, the low-temperature region which is concentrated on the central portion and the high-temperature region on the channel sidewalls. The temperatures distribution of MR-RC appears the highest among the other types due to the continuous growth of thermal boundary layer in stream-wise direction. The same trend of temperature distribution was observed in MC-SC, but the existence of cavities is relatively enhances heat transfer through increasing the surface area. Although, the stagnation zones within the cavities have a noticeable effect in deterioration of heat transfer due to inadequate flow mixing leading to rise in temperature. The existence of ribs in the central of channel has a significant effect in decreasing the temperature for both MC-RR and MC-SCRR. However, the flow disturbance which caused by a combination of ribs and cavities in MC-SCRR leads to more reduction in temperature than MC-RR. Fig. 10 shows the trends of average base surface temperature versus Reynolds number for all microchannel types. It is clearly shown that MR-RC has a highest average temperature followed by MC-SC, MC-RR and MC-SCRR respectively. The superiority of thermal performance of MC-SCRR can be attributed to many reasons:

- (1) The presence of ribs and cavities lead to increase heat transfer area that improves thermal performance. Fig. 11 illustrates the surface area for the four types, where MC-SCRR owns the largest area which rises about 6.6%, 14.1% and 20.5% more than MC-RR, MC-SC and MR-RC respectively.

- (2) The presence of ribs together with cavities urges to form vortices due to the pressure gradient which significantly enhance flow mixing and chaotic advection and therefore heat transfer.
- (3) Reduces the effect of stagnation zones at low velocity through guiding more flow towards cavities and promote disturbance.
- (4) The periodic interruption and redevelopment for thermal boundary layer besides the jetting and throttling effect are contributing in promoting heat transfer.

4.2.4. Performance analysis

The performance evaluation of proposed design is based on the comparison with the corresponding straight rectangular channel MC-RC. The Nusselt number ratio and friction factor ratio are the two parameters that used to identify the characteristics of thermal performance, hydrodynamic performance and overall performance.

Figs. 12 and 13 shows the variation of friction factor ratio (f/f_o) and Nusselt number ratio (Nu/Nu_o) with Reynolds number, where the subscript (o) refers to straight microchannel MC-RC. It can be observed that friction factor of MC-SC is the lowest and that of MC-RR is the highest. The enlargement in flow area, which caused by cavities, is contributing in reducing the friction losses. In regard to the thermal performance, Fig. 13 demonstrates the variation of Nusselt number ratio (Nu/Nu_o) versus Reynolds number. The MC-SC shows the lowest Nusselt number ratio due to stagnation zones effect. Also, it can be seen that MC-RR has the highest ratio for $Re < 400$, whereas the Nusselt number ratio of MC-SCRR shows superiority for $Re > 400$. This behaviour indicates that the merging of ribs and cavities is more effective at higher velocities due to its ability to induce high flow disturbances and formation of vortices which increase flow mixing.

The overall performance of the proposed design is measured using performance factor Pf which represents the Nusselt number ratio over friction factor ratio to the power of one third. The value of Pf indicates to extent of the thermal effectiveness relative to the pressure drop. The increment of Pf over 1 indicates to the superiority of thermal performance over pressure drop and vice versa if the value is less than one. Therefore, the design is considered acceptable if the value of Pf is more than one. The equation of Pf is given by:

$$Pf = \frac{Nu/Nu_o}{(f/f_o)^{1/3}} \quad (19)$$

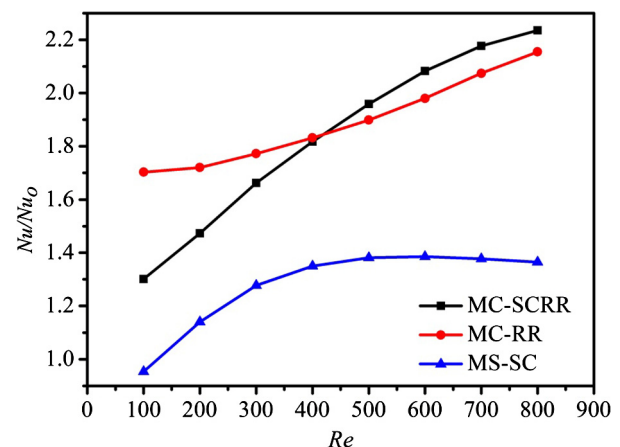


Fig. 13. Variation of Nusselt number ratio with Reynolds number for MC-SCRR, MC-RR and MC-SC.

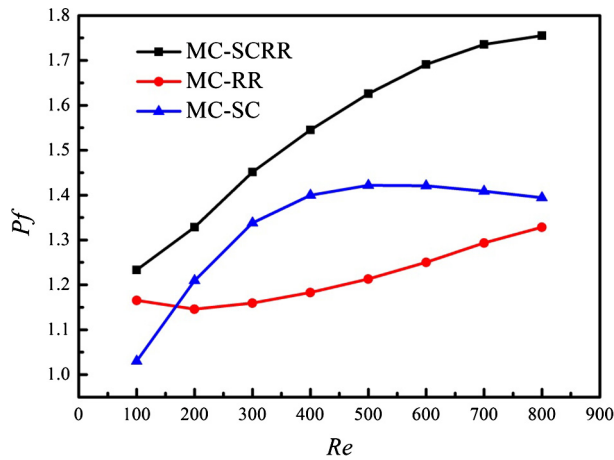


Fig. 14. Variation of performance factor Pf with Reynolds number for MC-SCRR, MC-RR and MC-SC.

Fig. 14 shows the variation of Pf with Reynolds number. It can be seen that Pf of all types are directly proportional with Reynolds number except in case of MC-SC where Pf has witnessed a slight decrease for $Re > 500$. The performance of MC-RR is better than that of MC-SC at low Reynolds < 150 due to the high flow disturbances which caused by ribs. However, with increasing in Reynolds number over 150, the performance of MC-SC predominates over MC-RR. Although the Nusselt number of MC-RR is higher than that of MC-SC, but the high pressure drop in MC-RR is significantly degrading the overall performance.

In general, the Pf of MC-SCRR is superior to both of MC-RR and MC-SC for $100 \leq Re \leq 800$. The maximum value of Pf for MC-SCRR is 1.755 at $Re = 800$, whereas the maximum value for MC-RR is 1.39 at $Re = 800$. The Pf of MC-SC increases when Reynolds number increases and persists to its maximum value of 1.421 at $Re = 500$ then reduces slightly to 1.394 at $Re = 800$.

4.3. The effect of geometrical parameters of rib and cavity on hydrothermal performance

According to the previous results, it can be concluded that MC-SCRR has achieved a highest performance compared with MC-RR and MC-SC. In order to identify the effects of geometrical parameters on the hydrothermal performance of MC-SCRR, three parameters have been chosen to be analysed; the relative amplitude of the cavity; the relative width of the rib and the relative length of the rib.

4.3.1. The effect of relative amplitude

The relative amplitude is defined as the ratio of cavity amplitude over cavity length:

$$\lambda = \frac{A}{L_c} \quad (20)$$

where (A) is the cavity amplitude and (L_c) is the cavity length. In order to explore the effect of relative amplitude, the cavity amplitude (A) has been changed to 0.025, 0.05, 0.075, 0.1 mm, whereas (L_c) is kept constant at 0.5 mm. So, the range of relative amplitude (λ) will be; 0.05, 0.1, 0.15 and 0.2. The relative rib length Γ and relative rib width β are fixed on 0.4 and 0.333 respectively as in Table 3.

Fig. 15 demonstrates the effect of relative amplitude cavity on the friction factor for range of Reynolds number $100 < Re < 800$. The friction factor decreases with the increase of Reynolds number for all cases. The blocking effect of the rib is more prominent at low ($\lambda = 0.05$) which leads to increase friction factor. This effect is

Table 3

The values of relative amplitude, relative rib width and relative rib length.

Relative cavity amplitude - λ	0.05	0.1	0.15	0.2
Relative rib width - β	0.333	0.333	0.333	0.333
Relative rib length - Γ	0.4	0.4	0.4	0.4

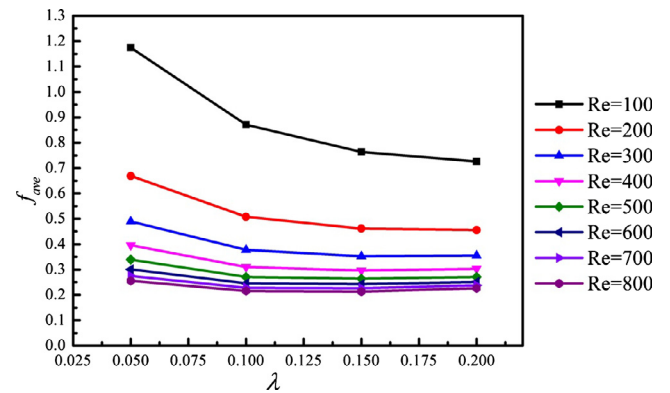


Fig. 15. Variation of average friction factor f_{ave} with relative amplitude λ for MC-SCRR at relative rib width $\beta = 0.333$ and relative rib length $\Gamma = 0.4$.

diminishes with the increasing of λ due to the flow area enlargement.

It can be seen that the microchannel with ($\lambda = 0.2$) achieved lowest value of friction losses at $Re = 100$ and 200. For $Re > 200$, the microchannel with ($\lambda = 0.15$) has showed a lowest values of friction factor.

Fig. 16 illustrates the effect of relative amplitude λ on average Nusselt number. The relative amplitude has a distinct effect on heat transfer enhancement. For all cases, the Nusselt number increases with the increase of Reynolds number. At $Re = 100$, the average Nusselt number of microchannel with $\lambda = 0.05$ achieved a highest value due to the flow disturbance which caused by ribs. For $Re > 100$, microchannel with the highest relative amplitude of ($\lambda = 0.2$) achieved the highest values of Nusselt number. The increase of λ leads to enlarging flow regions and surface area, promotes flow mixing due to the radial flow which forms vortexes besides the effect of jetting and throttling flow.

Fig. 17 shows variation of performance factor Pf with relative amplitude for different Re . It is found that performance factor Pf increases with both Reynolds number and relative amplitude. At Reynolds number between $100 < Re < 500$, the microchannel with relative amplitude of (0.2) shows the best performance factor. For Reynolds number higher than 500, the microchannel with rel-

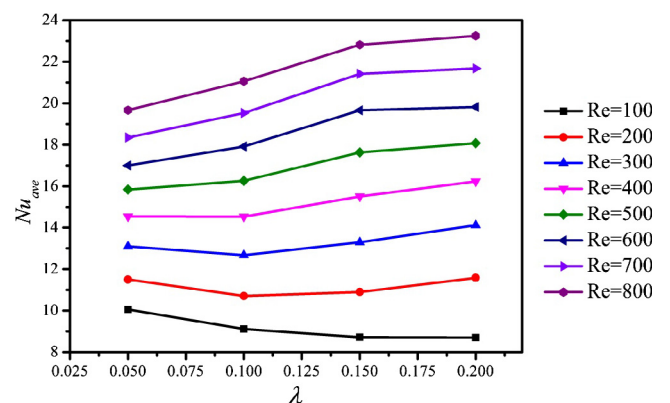


Fig. 16. Variation of average Nusselt number Nu_{ave} with relative amplitude λ for MC-SCRR at relative rib width $\beta = 0.333$ and relative rib length $\Gamma = 0.4$.

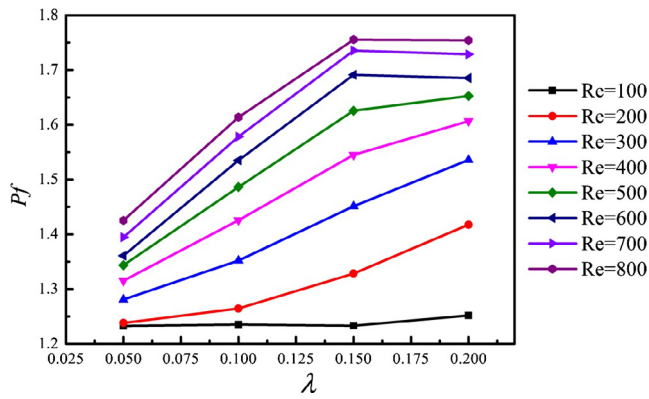


Fig. 17. Variation of performance factor Pf with relative amplitude λ for MC-SCRR at relative rib width $\beta = 0.333$ and relative rib length $\Gamma = 0.4$.

Table 4
The values of relative amplitude, relative rib width and relative rib length.

Relative rib width- β	0.2	0.3	0.4	0.5	0.6
Relative cavity amplitude - λ	0.15	0.15	0.15	0.15	0.15
Relative rib length- Γ	0.4	0.4	0.4	0.4	0.4

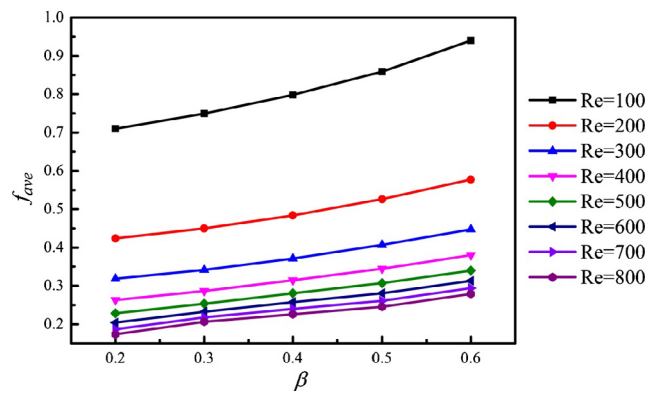


Fig. 18. Variation of average friction factor f_{ave} with relative rib width β for MC-SCRR at relative cavity amplitude $\lambda = 0.15$ and relative rib length $\Gamma = 0.4$.

ative amplitude of (0.15) shows the best performance factor. Although that relative amplitude of 0.2 provides larger flow area, but it also increases the vortex area which in turn increase friction factor and relatively degrades performance factor.

So, it can be concluded that relative amplitude of ($\lambda = 0.15$) can be adopted as an optimal relative amplitude for higher flow rate, while relative amplitude of ($\lambda = 0.2$) can be the optimal value for the moderate to low flow rate.

4.3.2. The effect of relative rib width

The relative rib width is defined as the ratio of rib width to the channel width as follows:

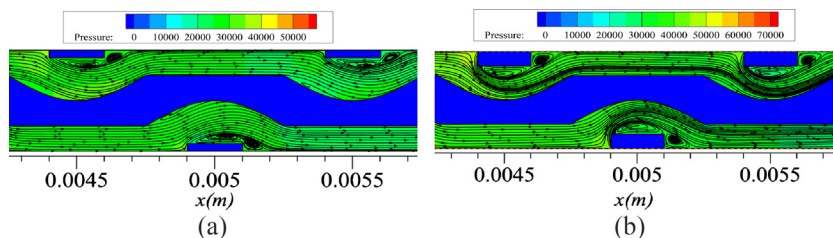


Fig. 19. Pressure distribution and streamlines in MC-SCRR at $Re = 600$ of (a) $\beta = 0.3$ (b) $\beta = 0.6$.

$$\beta = \frac{W_r}{W_c} \quad (21)$$

In order to study the influences of relative rib width, the relative amplitude λ and relative rib length Γ are fixed on 0.15 and 0.4 respectively. The relative rib width β increases according to Table 4.

Fig. 18 illustrates the influence of relative rib width on friction factor. It can be seen that the friction factor increases with increasing relative width which attributed to the increment of both blocking effect and flow recirculation. At the same time, it is found that friction factor decreases with increasing of Reynolds number. The friction factor with $\beta = 0.6$ has a highest value of 3.37 times higher than MC-RC for $Re = 800$. Fig. 19 shows the pressure distribution and streamlines in x-z plane for relative width ratio of $\beta = 0.3$ and 0.6 respectively. It can be clearly seen the differences in flow behaviour and pressure drop in both cases. In larger relative width of $\beta = 0.6$, intensive flow separation and flow recirculation are covered a large area on sides and rear zone of ribs with high pressure drop. In smaller relative width of $\beta = 0.3$, the flow recirculation and

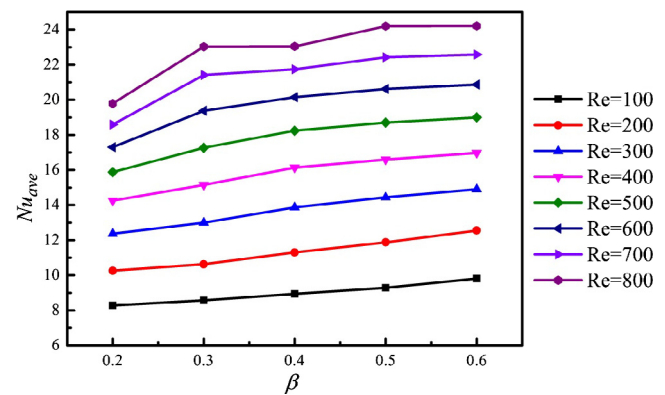


Fig. 20. Variation of average Nusselt number Nu_{ave} with relative rib width β for MC-SCRR at relative cavity amplitude $\lambda = 0.15$ and relative rib length $\Gamma = 0.4$.

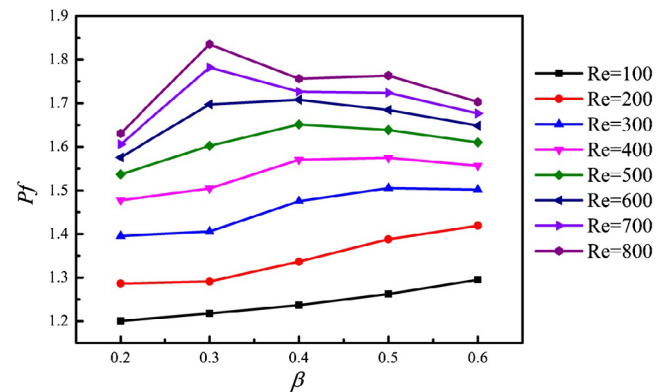
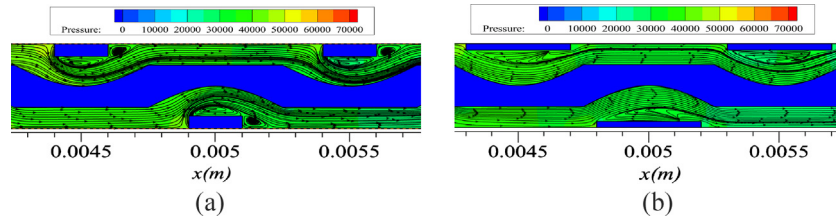


Fig. 21. Variation of performance factor Pf with relative rib width β for MC-SCRR at relative cavity amplitude $\lambda = 0.15$ and relative rib length $\Gamma = 0.4$.

Table 5

The values of relative amplitude, relative rib width and relative rib length.

Relative rib length - Γ	0.4	0.5	0.6	0.7	0.8
Relative cavity amplitude - λ	0.15	0.15	0.15	0.15	0.15
Relative rib width - β	0.3	0.3	0.3	0.3	0.3

**Fig. 22.** Pressure distribution and streamlines in MC-SCRR at $Re = 600$ of (a) $\Gamma = 0.4$ (b) $\Gamma = 0.8$.

separation is concentrated on small area behind ribs and on sides with lower pressure drop.

Fig. 20 shows the trend of Nusselt number versus relative rib width. It is clearly shown that Nusselt number increases with increasing of relative rib width and Reynolds number. The increment in relative rib width leads to intensify the vortexes, flow disturbances and flow separation which significantly contributing in heat transfer enhancement. The highest average Nusselt number has been achieved in microchannel with $\beta = 0.6$ which is 2.37 times greater than MC-RC for $Re = 800$.

Although of largest Nusselt number has been acquired at width ratio of $\beta = 0.6$, but the high friction factor led to deterioration in overall performance as illustrated in Fig. 21. It can be seen that maximum performance factor is deviated towards smaller rib width with increasing in Reynolds number due to high flow separation and recirculation. For $Re = 100$ and 200, the maximum performance factor Pf is found in microchannel with $\beta = 0.6$, while the maximum Pf is deviated to $\beta = 0.5$ at Re of 300 and 400. The same trend for β of 0.4, which has a maximum performance Pf at $Re = 500$ and 600. The best Pf has been achieved with $\beta = 0.3$ of $Pf = 1.835$ at $Re = 800$.

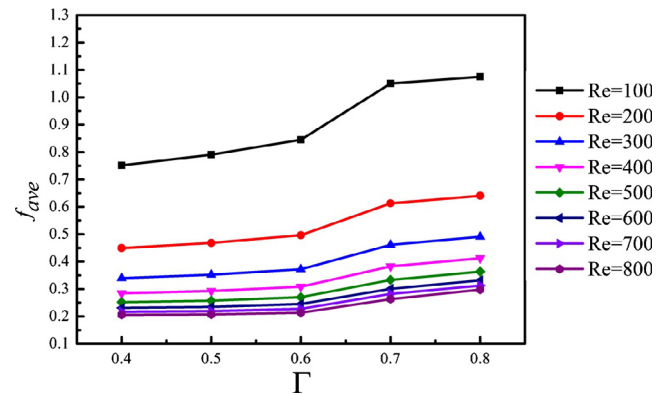
4.3.3. The effect of relative rib length

In order to identify the optimal rib length that produces the highest performance factor, the relative rib length parameter has been chosen to be analysed. The relative rib length is defined as the ratio of rib length over cavity length as follows:

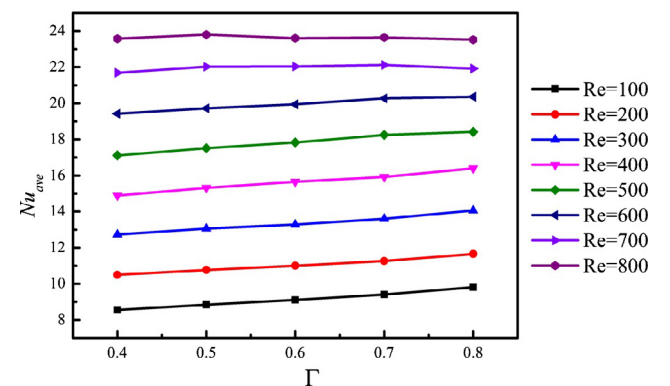
$$\Gamma = \frac{L_r}{L_c} \quad (22)$$

The relative cavity amplitude λ and relative rib width β are fixed on 0.15 and 0.3 respectively. The relative rib length is changed according to Table 5.

The increase in relative length causes a constrict in cavity inlet and outlet which substantially increase the vortexes and flow separation and thus increase friction factor. Fig. 22 illustrates two cases; long relative length of $\Gamma = 0.8$ and short relative rib length of $\Gamma = 0.4$. It can be clearly seen a large recirculation zone on the rib sides of $\Gamma = 0.8$ due to the constriction in inlet of cavity zone which leads to high pressure drop. Meanwhile, in short relative length of $\Gamma = 0.4$, the recirculation zone is shrunk to occupy a smaller area. The variation of friction factor with relative rib length is illustrated in Fig. 23. The friction factor of microchannel with $\Gamma = 0.8$ is the highest with 2.87 times more than MC-RC for $Re = 800$.

**Fig. 23.** Variation of average friction factor f_{ave} with relative rib length Γ for MC-SCRR at relative cavity amplitude $\lambda = 0.15$ and relative rib width $\beta = 0.3$.

The Nusselt number increases with increasing of Reynolds number for all values of relative width as in Fig. 24. A slight increase of Nusselt number has been observed with increasing in rib length for $Re < 700$ which attributed to the increment in surface area of heat transfer. In higher Reynolds numbers of 700 and 800, the increase in relative rib length leads to intensify the flow separation on the rib sides which consequently degrades relatively the heat transfer. So, it can be seen that Nusselt numbers are approximately equal for all rib lengths with a little progress at rib length of 0.5 which attains 2.3 times larger than MC-RC.

**Fig. 24.** Variation of average Nusselt number Nu_{ave} with relative rib length Γ for MC-SCRR at relative cavity amplitude $\lambda = 0.15$ and relative rib width $\beta = 0.3$.

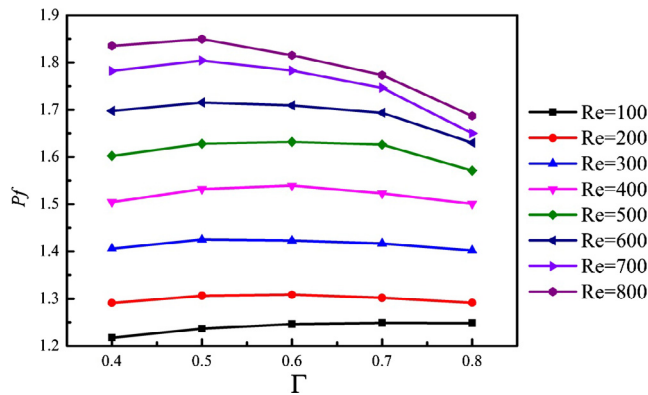


Fig. 25. Variation of performance factor Pf with relative rib length Γ for MC-SCRR at relative cavity amplitude $\lambda = 0.15$ and relative rib width $\beta = 0.3$.

The effect of relative rib length on performance factor for Reynolds number between 100 and 800 is shown in Fig. 25. In general, the performance factor increases with the increasing in Reynolds number. The rate of increasing in performance factor is not distinct for Reynolds number from 100 to 400. For $Re > 400$, the high friction factor which is accompanied with increasing of relative length contributes in retreating the rate of increasing in performance factor. The rate of increasing in performance factor is kept constant for moderate relative length as in $\Gamma = 0.5$ which earned the maximum value of $Pf = 1.85$ at $Re = 800$.

5. Conclusion

In present paper, the characteristics of fluid flow and heat transfer in microchannel heat sink with sinusoidal cavities and rectangular ribs MC-SCRR have been studied numerically. The combined effects of cavities and ribs are analysed through comparative with related geometries such as microchannel with rectangular ribs MC-RR and microchannel with sinusoidal cavities-SC. Furthermore, an optimization analysis has been performed to the geometrical parameters such as relative cavity amplitude λ , relative rib width β and relative rib length Γ to choose the optimal parameters that meet the cooling requirement of higher thermal performance in lower pressure drop. The following conclusions can be extracted from the results:

- (1) the using of hybrid techniques between heat transfer augmentation methods such as ribs and cavities can be significantly enhanced overall performance in microchannel heat sink rather than using individual technique.
- (2) The combination of ribs and cavities contributed in reducing the high pressure drop which caused by ribs in MC-RR through providing larger flow area. Additionally, this combination has contributed in eliminate the stagnation zone which existed in MC-SC through guiding the flow towards the cavity zone.
- (3) The proposed design promotes flow mixing and chaotic advection through the vortices formation in transverse direction of cavity.
- (4) The current design provides larger heat transfer area comparative with relevant microchannel geometries such as MC-RC, MC-RR and MC-SC.
- (5) The optimization process of geometrical parameters of relative cavity amplitude, relative rib width and relative rib length shows that:
 - (a) The Nusselt number increase as the relative cavity increases, whereas the friction factor shows opposite trend.

- (b) The Nusselt number and friction factor increase as the relative rib width increases.
- (c) The Nusselt number and friction factor increase as the relative rib length increases.
- (d) The MC-SCRR with parameters of relative cavity amplitude with $\lambda = 0.15$, relative rib width $\beta = 0.3$ and relative rib length $\Gamma = 0.5$ yields the best overall performance with $Pf = 1.85$ at $Re = 800$.

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